

LET'S SOLVE A GUESSTIMATE





Estimate the market size for air-conditioners in Mumbai.

PRELIMINARY QUESTIONS

Interviewee- There are two types of air-conditioners- windows and split. Shall I include both?

Interviewer - Yes.

Interviewee- And should I include objects like air-coolers?

Interviewer - No, you need not

Interviewee- In what unit would you want me to estimate the market size- #ACs, INR, etc.?

Interviewer - Good question. Estimate the market size in terms of tonnage. A 1 ton of AC can typically cool a small room.

Interviewer - Interesting, but no. Include only first-hand ACs in the market size

Interviewee- Air Conditioners are used for residential cooling and commercial cooling in Mumbai. Residential cooling involves cooling in homes and apartments. Commercial cooling would involve Office Spaces, Malls, Restaurants, Hospitals, Schools, etc. Shall I calculate the market size including all these?

Interviewer - For simplicity, just calculate Residential and Office Space cooling. We can neglect the rest.

APPROACH TO USE

We need to calculate the market size for air-conditioners in tons. The market size would involve the #ACs required to replace old ones along with the additional demand for new ACs.

Market size in tons = Demand to replace old ACs + Demand due to growth in market size
= (#ACs in tons) / (Avg. life of an AC) + (#ACs in tons) x (Growth rate)

Avg. Replacement Demand = Current # Products / Avg. life cycle of product

If the life cycle of a product is Y years on average, then in Y years' time each of these products would be replaced by newer ones. Then we can say, Average replacement demand for any particular year is (Current # Products/Y).

Interviewee- We can take the average life of an AC to be 10 years. Can we take the growth rate in absence of any other information as the growth in GDP of India which is about 5%?

Interviewer - Take the growth rate to be 10%

Interviewee- All right, what remains to be calculated is the (#ACs in tons) presently in Mumbai. We are looking at demand in

- Residential Segment
- Office Space Segment

Now, for the Residential segment,

**#ACs in tons = (#Families who can afford ACs)
x (Avg tons of ACs per family)**

Interviewee- For the Office Space Segment, Offices typically have centralized ACs and the tonnage of the Central AC decides how much area can be cooled.

#ACs in tons = (Total office area) x (Tons required per unit area)

= (#People working in air-conditioned offices) x (Office area per person) x (Tons required per unit area)

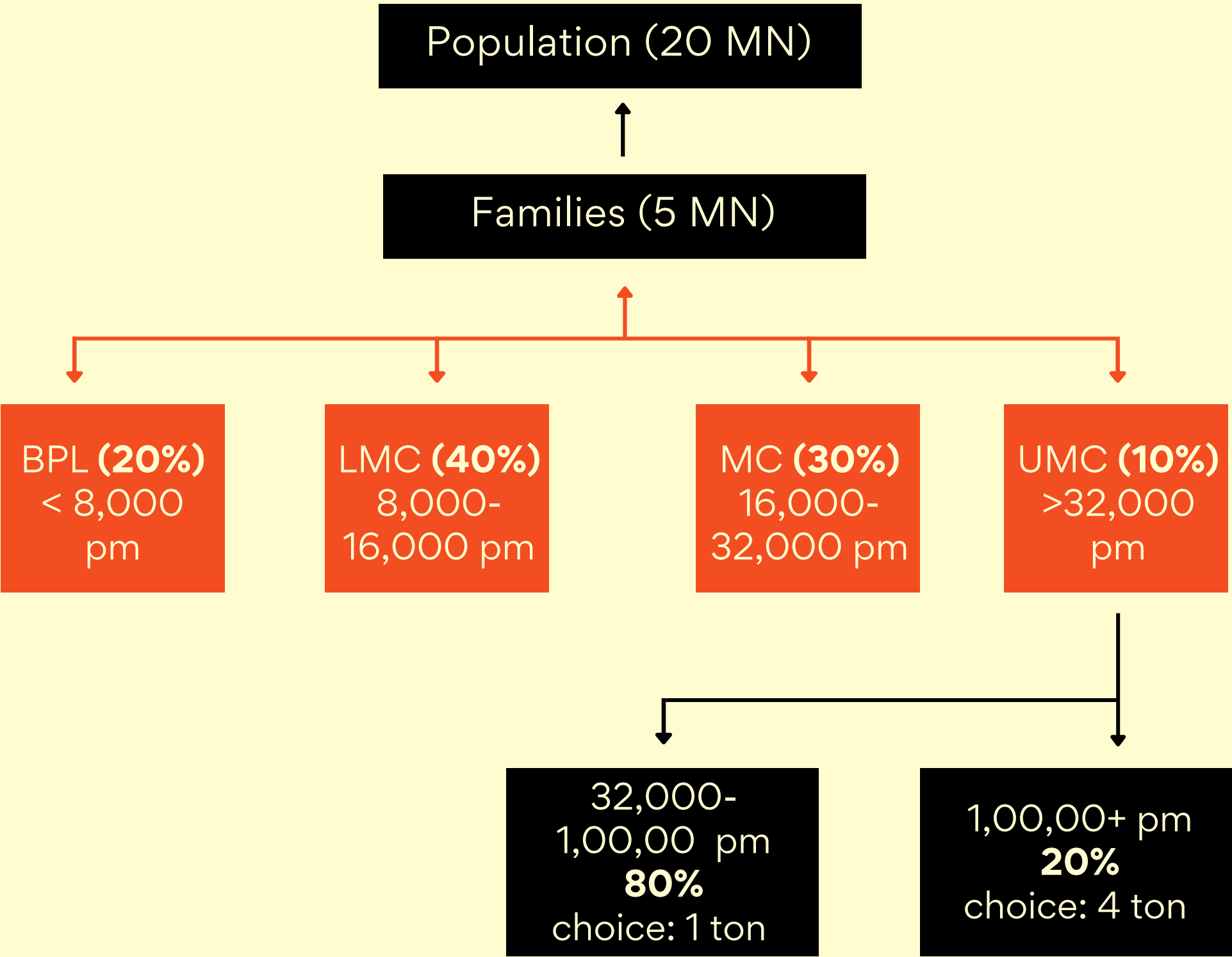
Interviewer - Sounds good. Go ahead

RESIDENTIAL SEGMENT

Interviewee- Taking an average family size of 4 in Mumbai for simplification. Hence,
= 5 MN families for 20 MN population

Next, we are looking at the affordability of an Air Conditioner. An AC costs roughly INR 25,000 to 40,000 depending on the tons.

A Middle-Class family has an income in the range of 16,000 to 32,000 pm. The average household income would be around 20,000pm. Given the AC cost is more than a month's salary, I do not think that the Middle Class would be able to afford ACs. The Below Poverty Line (BPL) and Lower Middle Class (LMC) families would not be able to afford it as well.



Abbreviation Support: MN- million; pm- per month; LMC- Lower Middle Class; MC- Middle Class; UMC- Upper Middle Class.
Kindly note all salaries mentioned here are in INR.



Interviewee- Now we need to see the Upper Middle Class (UMC) families. Their income is greater than 32,000 pm. The average income of this segment would be roughly 50,000 pm or about 6 lakhs per annum. The average UMC family would live in a 2 BHK home. I think such a family would be able to afford at least a 1 ton of AC

In this segment there would be families which can even afford up to 3ACs, two 1 ton ACs for each bedroom, and a 2 ton AC for the living room. This is the case for my family, and our average household income would be around 100,000pm. Based on this we can split the UMC into two segments.

Interviewee- The segments would be:

- **Income between 32,000pm-100,000pm:** Let's assume this will be 80% of the UMC segment. We will assume on average a family in this segment owns a 1-ton AC.
- **Income greater than 100,000pm:** Based on our assumption this will be 20% of the UMC segment. We will assume the average family in this segment owns 4 tons of AC.

Hence we get,

#Tons in residential segment= 5MN x (10%) x (80%) x (1 ton) + 5MN x (10%) x (20%) x (4 tons)

=400,000 + 400,000

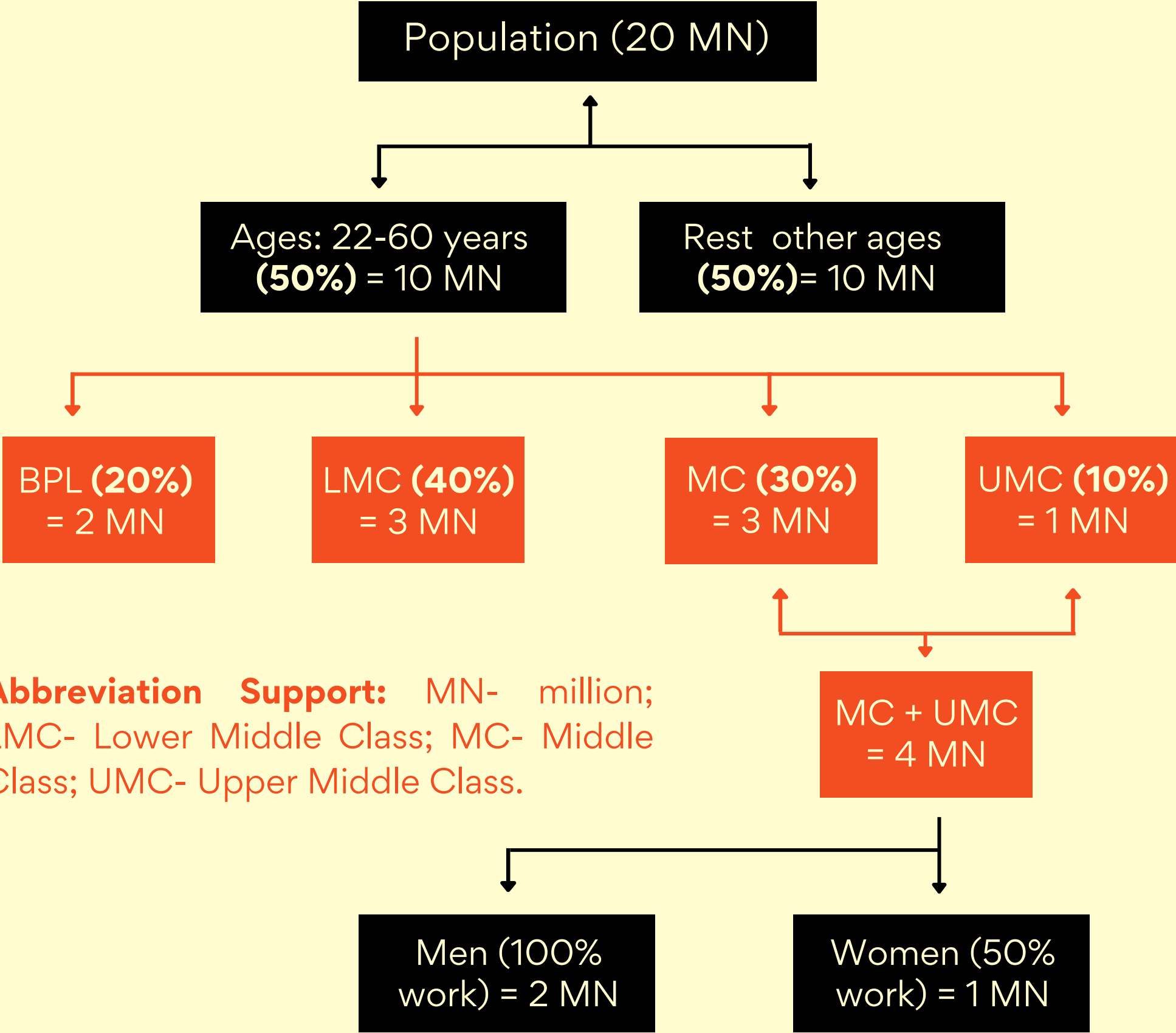
=800,000 tons.

OFFICE SPACE SEGMENT

Interviewee- For the Office Space Segment
#ACs in tons= (#People working in air-conditioned offices) x (Office area per person) x (Tons required per unit area)

The people working in air-conditioned offices typically would have white-collar jobs. Though there will be cases where white-collar professionals do not work under ACs and blue-collar workers have air conditioning, we will neglect these exceptions.

Please Note: (White-collar employees are those who typically work in offices, they are accountants, bankers, bureaucrats, etc. Blue-collar employees would-be mechanics, electricians, manual laborers, etc.)



Abbreviation Support: MN- million;
LMC- Lower Middle Class; MC- Middle
Class; UMC- Upper Middle Class.

Kindly note that we're assuming the MC+ UMC
population to be divided as follows:
Men (50%)= 2 MN; Women (50%)= 2 MN



Interviewee- The people with white-collar jobs would be part of the Middle Class and Upper Middle Class in the age group of 22-60. Assuming an equal spread of the population from 1-80 yrs., roughly 50% of the population will lie in the 22-60 bracket.

We can assume that all men and half the women would be working. **This gives us a total of 2MN men + 1MN women = 3MN people.**

In a typical office, people work in cubicles. An air-conditioned office has several common areas such as receptions, washrooms, and lunch areas. If we were to divide the total area of an office by the total number of employees, we can say that an office area roughly the same as a 'bedroom' would be equivalent to two employees

Interviewee- A 'bedroom' typically has an area of 200 sq. ft. Hence each employee is equivalent to a 100 sq. ft. area in an office. Also, we will require a 1-ton AC to cool a room of 200 sq. ft.

Hence,

$$\begin{aligned}\text{\#ACs in tons} &= (\text{\#People working in air-conditioned offices}) \times (\text{Office area per person}) \times (\text{Tons required per unit area}) \\ &= (3\text{MN}) \times (100 \text{ sq. ft.}) \times (1 \text{ ton} / 200 \text{ sq. ft.}) \\ &= \mathbf{1.5\text{MN tons}}\end{aligned}$$

Now,

$$\begin{aligned}\text{Total \#ACs} &= \text{ACs in Residential Segment} + \text{ACs in Office Space Segment} \\ &= 800,000 \text{ tons} + 1.5 \text{ MN tons} \\ &= \mathbf{2.3 \text{ MN tons.}}\end{aligned}$$

Interviewee- Hence,

Market Size in tons = (#ACs in tons) / (Avg. life of an AC) + (#ACs in tons) x (Growth rate)
= (2.3MN tons) / (10 yrs.) + (2.3MN tons) x (10% per year)

= 460,000 tons / year

Interviewer - Great. We can close the case here.

That's a Wrap!

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